

6. You are exposed to natural sources of background radiation all the time.  
The level of background radiation varies depending on where you live.

(a) (i) Name **one** cause of background radiation. [1]

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(ii) Give a reason why the level of background radiation depends on where you live. [1]

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(b) Having an X-ray taken of part of the body exposes the person to ionising radiation.  
The following table gives the effect on the body from X-rays.

| X-ray  | Radiation's effect on the body |                                                    |
|--------|--------------------------------|----------------------------------------------------|
|        | Radiation dose (units)         | Equivalent number of hours of background radiation |
| chest  | 100                            | 200                                                |
| hand   | 1                              | 2                                                  |
| dental | 5                              | 10                                                 |

(i) State which X-ray gives the most risk to the patient and give a reason for your answer. [2]

X-ray .....

Reason .....

.....

(ii) A person has 1 chest X-ray and 4 dental X-rays in his lifetime.  
Toni suggests that this is equivalent to 240 hours of extra background radiation.  
Use data from the table to decide whether you agree with this suggestion. [3]  
Space for calculation.

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